

Image Based Hand Tracking via Interacting Multiple Model and Probabilistic Data Association (IMM-PDA) Algorithm

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Abstract

Traditional image based hand tracking uses a single Kalman filter to estimate and predict the hand state (position, velocity, and acceleration). However, this approach may fail in the case of large maneuvers and cluttered measurements. In this paper we propose to use the interacting multiple model (IMM) filter to catch a maneuver and the probabilistic data association (PDA) method to process noisy measurements and false alarms. A theoretical framework of image based hand tracking by IMM-PDA is set up. Experiment results from several video segments show that IMM-PDA can successfully track hand motions in a natural conversational environment.

1. Introduction

Accurate estimation and prediction a hand state is a very important issue in human gesture study, robots navigation, computer vision and pattern recognition. A good estimation and prediction can not only improve tracking performance, but also save computation time.

To estimate and predict an object state in image based tracking, several methods, i.e., color intersection method [1], pseudo hidden Markov model [2], Kalman filter (KF) [3], wavelet network [4], as well as probability detection [5], have been proposed during the last decade. Among these methods, KF has been extensively used in tracking human bodies [6, 7, 8], faces [9, 10, 11] as well as hands [12, 13, 14, 15].

It is well known that to directly use KF for estimation and prediction, one needs to correctly model the system and provide a measurement at each instance. However, because of the following two facts, the single KF model approach cannot accurately estimate or predict the hand state in a natural conversational environment.

- *Unknown maneuver.* Due to the existence of an unknown maneuver, we cannot obtain the exact system equation(s) to describe the dynamical hand motion.
- *Multiple noisy measurements.* Because of the influences of other objects¹ and the cluttered environment, the measurements consist of two types of data. i.e., the correct measurement (if the hand can be detected and it falls in the validation gate) and the incorrect measurements (originating from ambiguity of the image or false alarms). Since the sensor(s) cannot distinguish the correct and incorrect measurements, we have multiple measurements at each instance.

For the system modelling side, there are three approaches to approximately simulate the unknown maneuvers [16]. The first approach, *random process approximation*, treats the unknown maneuver term as an additional noise with different magnitudes at different times. The second approach, *input estimation*, assumes the maneuver to be constant over a certain period of time and the estimation can be done based on the least square criterion. The third approach is named *interacting multiple model* (IMM) which assumes that (i) the system behaves according to one of a finite number of models, (ii) each model has its own noise level and structure, and (iii) the system state can be obtained via combining results of different models with different probabilities.

For the measurement side, the data association technique can be used to process noisy multiple measurements and false alarms at each instance. The main idea of a data association technique can be briefly summarized as follows. Suppose both the system and measurement are driven by Gaussian noise, and the estimated hand state and its covariance at time $k - 1$ is known. Then, according to the system equation(s), we can predict the measurement state and its

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¹For example, the interlocutor's hands, the heads of both the subject and interlocutor, and the third person's hands and head et al.

associated covariance matrix at time k . Assuming that the true measurement at time k , conditioned on the measurement history, is normally distributed, one can select the candidate measurement(s) according to the probability value of each measurement with a particular threshold. To do so, several data association approaches, such as the *nearest-neighbor standard filter* [17], the *track-splitting filter* [18], the *probabilistic data association filter* (PDA) [19], the *optimal Bayesian filter* [20], the *Joint Probabilistic Data Association approach* [21], the *multiple hypothesis tracking* [22], and the *interacting multipattern probability and joint probability data association* [23], have been proposed in the past three decades.

Among these techniques, IMM algorithm has demonstrated its ability to catch up the unknown maneuver, while the PDA data association approach has shown its relative easy-implementation, and good performance properties in the object tracking community in recently years. Hence, we prefer to combine them together, i.e., using IMM-PDA to track human hand in the natural conversation environment in this paper. Further, in order to evaluate the performance of the IMM-PDA algorithm, we also compare the IMM-PDA results with those from single Kalman filter models. The experiment results show that the IMM-PDA algorithm gives the best performance.

The rest of the paper is organized as follows. In section 2, we formulate the hand tracking problem under single KF model, and IMM-PDA framework. Three tracking filters, their parameters and initializations are presented in section 3. The experiment setup, the measurements and the tracking results of the three filters are introduced in section 4. A summary and discussion is given in section 5.

2. Problem formulation

2.1. Single KF approach

In this paper, we use $\underline{x}(k)$ to describe the hand state at frame k , which is expressed as

$$\underline{x}(k) = [x, v_x, a_x, y, v_y, a_y]^T, \quad (1)$$

where x, y, v_x, v_y, a_x and a_y stand for the positions, velocities and accelerations in x and y directions, respectively.

Since the hand motion obeys the Newton law, the relationship of hand states between time k and $k + 1$ can be formulated as

$$\underline{x}(k + 1) = \mathbf{F}(k)\underline{x}(k) + \mathbf{G}(k)\underline{u}(k) + \underline{v}(k) \quad (2)$$

where $\mathbf{F}(k)$ is a 6×6 matrix which describes the system evolution behavior, $\underline{u}(k)$ is the unknown input, $\mathbf{G}(k)$ is the

coefficient matrix of $\underline{u}(k)$, and $\underline{v}(k)$ is system noise. Suppose $\underline{v}(k)$ is Gaussian white with the properties

$$E\{\underline{v}(k)\} = 0, \quad E\{\underline{v}(k_1)\underline{v}^T(k_2)\} = \mathbf{Q}(k_1)\delta_{k_1k_2}, \quad (3)$$

where $\mathbf{Q}$ is the covariance matrix, and $\delta_{ij} = 1$ for $i = j$, $\delta_{ij} = 0$ otherwise.

On the measurement side, suppose we can only measure the center position of the hand at each frame. Then, the measurement state $\underline{z}(k)$, is determined by ²

$$\underline{z}(k) = \mathbf{H}(k)\underline{x}(k) + \underline{w}(k), \quad (4)$$

where $\underline{z}(k) = [x(k), y(k)]^T$, $\mathbf{H}$ is the transform matrix, which can be written as

$$\mathbf{H}(k) = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \end{bmatrix},$$

and $\underline{w}(k)$ is the measurement noise. Further, we assume $\underline{w}(k)$ is Gaussian white and there is no correlation with system noise $\underline{v}(k)$, i.e.,

$$\begin{aligned} E\{\underline{w}(k)\} &= 0, & E\{\underline{w}(k_1)\underline{w}^T(k_2)\} &= \mathbf{R}(k_1)\delta_{k_1k_2}, \\ E\{\underline{w}(k_1)\underline{v}^T(k_2)\} &= 0, \end{aligned} \quad (5)$$

where $\mathbf{R}$ can be simplified as

$$\mathbf{R}(k) = \text{diag}\{(\sigma_z^{(x)})^2, (\sigma_z^{(y)})^2\}.$$

Suppose the measurement at time k is $\underline{z}(k)$, and the cumulative set of measurements up to time k is

$$\mathbf{Z}^k = \{\underline{z}(j)\}_{j=1}^k. \quad (6)$$

If we know (2), (4) and (6), under the framework of Kalman filter[3], the estimated hand state

$$\hat{\underline{x}}(k|k) = E[\underline{x}|\mathbf{Z}^k], \quad (7)$$

and its covariance

$$\mathbf{P}(k|k) = E\{[\underline{x}(k) - \hat{\underline{x}}(k|k)][\underline{x}(k) - \hat{\underline{x}}(k|k)]^T\}, \quad (8)$$

at time k can be recursively computed. Further one can predict hand states in the next several frames by propagating $\hat{\underline{x}}(k|k)$ according to (2). This is the so called single KF model approach.

2.2. IMM approach

Since the maneuver is unknown, i.e., we do not know term $\mathbf{G}(k)\underline{u}(k)$ in (2), usually the single KF model can not give good performance. However, as mentioned in section 1, we can use a multiple model approach to approximate the maneuver. The IMM approach assumes that the system behavior can be simulated by the combination of a finite number of models, each of which has its own noise level and structure. Considering the characteristics of hand dynamics and computational time consuming, four models, i.e., a stop

²In the case of both position and velocity of a hand can be measured, the same equation as (4) can be used to describe the measurement system, but with different $\mathbf{H}$ and $\underline{w}$.

model, a constant velocity model(CV), a constant acceleration model with small noise(CA1), and a constant acceleration model with large noise(CA2) are used to describe the hand motion. Under this framework, system equation (2) becomes

$\underline{x}_i(k+1) = \mathbf{F}_i(k)\underline{x}_i(k) + \underline{v}_i(k), i = I, II, III, IV,$ (9)
where $\underline{v}_i(k)$ is Gaussian white with zero mean and covariance $\mathbf{Q}_i$. The system state at time k is a combination of different models, i.e.,

$$\hat{\underline{x}}(k) = \sum_{i=1}^{N_r} \mu_i(k) \hat{\underline{x}}_i(k), \quad (10)$$

where $\mu_i(k)$ is the probability of the i th model, and $\hat{\underline{x}}_i(k)$ is the estimated state from i th model with the current measurements. The details of the four models are introduced as follows.

(i) **model I (M_I): Stop model.** To catch the stop state of hand motion, the stop model is selected, which can be described by

$$\mathbf{F}_I = \begin{bmatrix} \mathbf{F}_{stop} & \mathbf{O}_{33} \\ \mathbf{O}_{33} & \mathbf{F}_{stop} \end{bmatrix}, \mathbf{Q}_I = \sigma_I^2 \begin{bmatrix} \mathbf{Q}_{stop} & \mathbf{O}_{33} \\ \mathbf{O}_{33} & \mathbf{Q}_{stop} \end{bmatrix}, \quad (11)$$

where

$$\mathbf{F}_{stop} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \mathbf{Q}_{stop} = \begin{bmatrix} \frac{1}{4}T^4 & 0 & 0 \\ 0 & T^2 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad (12)$$

and $\mathbf{O}_{33}$ is a 3×3 zeros matrix.

(ii) **model II (M_{II}): CV model.** Considering the discrete-time model, the system transition matrix and the covariance of $\underline{v}_{II}$ can be written as

$$\mathbf{F}_{II} = \begin{bmatrix} \mathbf{F}_{CV} & \mathbf{O}_{33} \\ \mathbf{O}_{33} & \mathbf{F}_{CV} \end{bmatrix}, \mathbf{Q}_{II} = \sigma_{II}^2 \begin{bmatrix} \mathbf{Q}_{CV} & \mathbf{O}_{33} \\ \mathbf{O}_{33} & \mathbf{Q}_{CV} \end{bmatrix}, \quad (13)$$

where

$$\mathbf{F}_{CV} = \begin{bmatrix} 1 & T & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \mathbf{Q}_{CV} = \begin{bmatrix} \frac{T^4}{4} & \frac{T^3}{2} & 0 \\ \frac{T^3}{2} & T^2 & 0 \\ 0 & 0 & 0 \end{bmatrix}. \quad (14)$$

(iii) **model III (M_{III}): CA with small noise intensity.**

Using the discrete-time CA model to describe the noise intensity, the CA model can be expressed as

$$\mathbf{F}_{III} = \begin{bmatrix} \mathbf{F}_{CA} & \mathbf{O}_{33} \\ \mathbf{O}_{33} & \mathbf{F}_{CA} \end{bmatrix}, \mathbf{Q}_{III} = \sigma_{III}^2 \begin{bmatrix} \mathbf{Q}_{CA} & \mathbf{O}_{33} \\ \mathbf{O}_{33} & \mathbf{Q}_{CA} \end{bmatrix}, \quad (15)$$

where

$$\mathbf{F}_{CA} = \begin{bmatrix} 1 & T & \frac{T^2}{2} \\ 0 & 1 & T \\ 0 & 0 & 1 \end{bmatrix}, \mathbf{Q}_{CA} = \begin{bmatrix} \frac{1}{4}T^4 & \frac{1}{2}T^3 & \frac{1}{2}T^2 \\ \frac{1}{2}T^3 & T^2 & T \\ \frac{1}{2}T^3 & T & 1 \end{bmatrix}. \quad (16)$$

(iv) **model IV (M_{IV}): CA with large noise model.**

This model is the same as M_{III} but it has big noise intensity described by σ_{IV} .

2.3. Probabilistic data association(PDA)

For the measurement side, because of image noise and false alarms, we may have n_k measurements at time k , i.e.,

$$Z(k) = \{z_i(k)\}_{i=1}^{n_k}. \quad (17)$$

If $n_k = 0$ (no detections is reported by the sensor), then $Z(k)$ is empty. The cumulative set of measurements up to time k in (6) becomes

$$Z^k = \{Z(j)\}_{j=1}^k. \quad (18)$$

The main idea of PDA can be briefly summarized as follows. Suppose the estimated hand state and its covariance at time $k-1$ is $\hat{\underline{x}}(k-1|k-1)$ and $\mathbf{P}(k-1|k-1)$, respectively. Then, according to (4), the predicted value of the measurement state at time k is $\hat{z}(k|k-1)$; the associated covariance matrix $\mathbf{S}(k|k-1)$ can be computed by the KF algorithm. Assuming that the true measurement at time k , conditioned on Z^{k-1} , is normally distributed, one can select the validation $m_k(\geq 0)$ measurement(s) according to the probability value of each measurement $z_j(k), j = 1, \dots, n_k$. Then, we run the m_k KFs parallelly, and the final state can be updated via the combination of all the m_k filters with different probabilities.

3. Tracking filters and their parameter selections and initializations

To compare the performance of the IMM-PDA approach, we implement the following three filters³.

(i) **Application of CV model.** The system equation of CV model is described by (9), (13) and (14), σ_{CV} is used to describe its noise intensity.

(ii) **Application of CA model.** The system equation of CA model is described by (9), (15) and (16), σ_{CA} is used to describe its noise intensity.

(iii) **Application of IMM model.** As mentioned in section 2.2, the IMM approach is a combination of a stop model, a CV model, and two CA models. the noise intensities of these models are $\sigma_I, \sigma_{II}, \sigma_{III}, \sigma_{IV}$. To describe Markov property of the transition from one model to another, we use p to denote the model probability transition matrix.

In our experiments, the noise intensities are selected as $\sigma_{CV} = 1000, \sigma_{CA} = 100, \sigma_I = 10, \sigma_{II} = 50, \sigma_{III} =$

³Because of the page limitation, we do not present the single KF-PDA and IMM-PDA implementation algorithms. The interested reader can contact the first author to request the details of the implementation algorithms and matlab programs.

1000, $\sigma_{IV} = 1500$, and $p = p_1$ if there is no measurement, $p = p_2$ if there are some measurements, where

$$p_1 = \begin{bmatrix} .7 & .1 & .1 & .1 \\ .24 & .5 & .13 & .13 \\ .24 & .13 & .5 & .13 \\ .24 & .13 & .13 & .5 \end{bmatrix}, p_2 = \begin{bmatrix} .5 & .1 & .3 & .1 \\ .05 & .7 & .15 & .1 \\ .05 & .1 & .7 & .15 \\ .05 & .1 & .15 & .7 \end{bmatrix}$$

Moreover, for all the three filters, the PDA approach is used to associate the measurements. To implement the PDA algorithm, besides the dimension related parameters, one needs to set the following parameters: (i) the detection probability P_D , (ii) the gate probability P_G , and (iii) the threshold of gating γ . In this work we take $P_D = 0.99$, $P_G = 0.99$, and $\gamma = 9.21$. Further, the covariance of measurement noise $\mathbf{R}$ is set as $\mathbf{R} = \text{diag}\{900, 625\}$ ⁴

The initial states of all the three filters are taken as

$$\hat{\mathbf{x}}(0|0) = [\bar{x}, 0, 0, \bar{y}, 0, 0]^T, \quad (19)$$

while their corresponding covariance matrices are selected as

$$\mathbf{P}(0|0) = \text{diag}\{\sigma_x^2 \mathbf{Q}_0, \sigma_y^2 \mathbf{Q}_0\}. \quad (20)$$

Where

$$\bar{x} = \frac{1}{n_x} \sum_{j=1}^{n_x} z_j^x(1), \quad \bar{y} = \frac{1}{n_x} \sum_{j=1}^{n_x} z_j^y(1),$$

$$\sigma_x^2 = \frac{1}{n_x} \sum_{j=1}^{n_x} (z_j^x(1) - \bar{x})^2, \quad \sigma_y^2 = \frac{1}{n_x} \sum_{j=1}^{n_x} (z_j^y(1) - \bar{y})^2,$$

$\mathbf{Q}_0 = \text{diag}\{1, 1/T, 1/T^2\}$, and n_x is the number of the total measurements at the first frame.

4. Experiment results

To study the human gesture behavior in the natural conversational environment, cooperating with psychologists and doctors, VISLab⁵ has collected about 100 experiments. As shown in Figure 1, Each experiment typically involves either two subjects discussing a spatial action plan or a subject either describing his/her living space or conveying the contents of a set of cartoon video clips to an interlocutor. A set of 3 to 5 cameras (a stereo pair trained at each participant with one camera zoomed in the head of the main subject) are used to collect the video streams. The sampling frequency of the videos is 30Hz, and the videos are typically 10 to 15 minutes in length.

The *vector coherence mapping* (VCM)[24] algorithm is used to measure the dynamical hand motion. However, since the following scenarios always happen in the natural conversational environment, the VCM results are very noisy and contain false alarms. (i) Since the non-focused

⁴In our experiment, the typical hand size is about 60×50 pixels. we assume the distance between the measured hand center position and the true center is about 30 pixels in x direction and 25 pixels in y direction.

⁵The Vision Interfaces and Systems Laboratory, Wright State University



Figure 1: The typical scene captured from the left camera of our experiment lr-31-t-c1.dv.

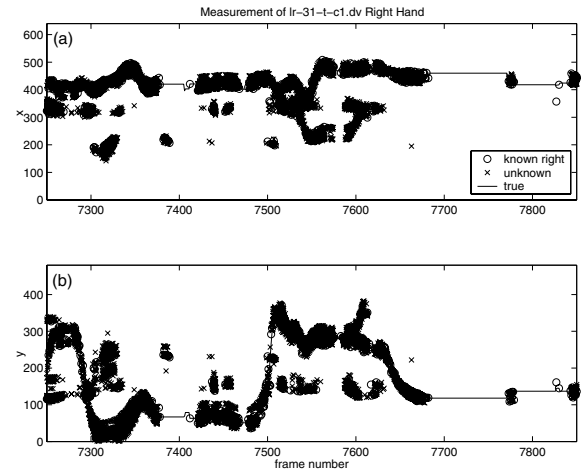


Figure 2: A segment of the measured center position of the rough clusters from the right hand in video lr-31-t-c1.

hands/heads may move during the conversation, these objects are caught by VCM. This forms the false alarms. (ii) When no object is moving, there is no measurement reported from VCM. (iii) When the two hands are overlapped or joined with each other, VCM can not distinguish them with each other. (iv) The existence of image noise. Figure 2 shows a segment of the center position of rough clusters from the right hand in our experiment LR-31-c1.

To compare the tracking results from different filters, we plot the time series of the true⁶ and the estimated hand positions in Figure 3. Moreover, to understand the mechanism of IMM, we also plot the number of validation measurements (m_k) inside the gate and the model-selection-

⁶The true positions are collected by the following three steps: (i) run VCM to get noisy measurements, (ii) run VCM-Cluster-Fixer[25] to approximate estimate the next 300 frames based on current position, (iii) Visualizing, verifying, and modifying the estimated results frame by frame by eye guiding via VCM-Cluster-Fixer.

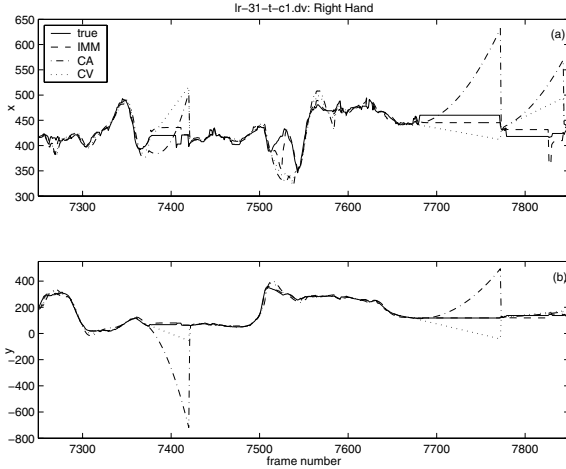


Figure 3: The estimated and true positions from the three application filters.

Table 1: The maximum and average absolute errors.

Filter	x	v_x	a_x	y	v_y	a_y
IMM(max)	67	720	27900	37	579	24236
(mean)	8	60	2401	5	54	1891
CA(max)	192	888	27739	779	938	24170
(mean)	27	100	2422	53	149	1942
CV(max)	119	790	27900	166	713	23236
(mean)	20	77	2392	25	88	1882

probabilities as a function of frame number in Figure 4. These figures show that when there has a big maneuver(for example, at frame number $k = 7316$ and $k = 7690$ the hand stopped immediately from the moving state), single KF models can not catch it and the estimated results are far away from the true positions. However, the IMM can catch the maneuver very well. The selected-model-probabilities (Figure 4(b) to (e)) also indicate that the stop model becomes the domain model when the hand is stopped. Further, we also compute the maximum and average absolute state errors of the three filters. The results are shown in Table 1. Other experiment results (not shown) also demonstrate that (i) in the near zero acceleration frame ranges the $M_{II}(\text{CV})$ is selected in the high probability, (ii) in the small acceleration frame ranges, the $M_{II}(\text{CA with small noise})$ is the domain model, (iii) in the big acceleration frame ranges the $M_{IV}(\text{CA with big noise})$ is selected in the high probability, and (iv) the IMM-PDA approach gives the best performance among the three filters.

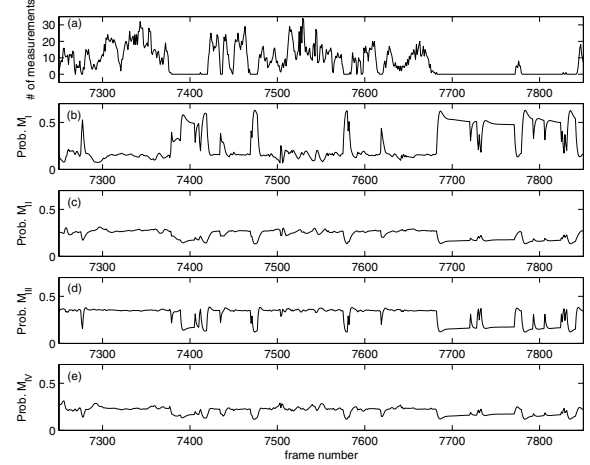


Figure 4: (a): The number of measurements inside the gate from IMM-PDA filter. (b) to (e): The dynamical behavior of model probabilities for stop, CV, CA1, and CA2 respectively.

5. Conclusion and future work

In this paper, we review the main challenges. i.e., the unknown maneuver and multiple noise measurements, in the image based hand tracking under the framework of object(s) tracking approach. Then we propose to use the IMM algorithm to take the unknown maneuver and PDA to process the multiple noise measurements. The problem formulations of both the single KF-PDA and IMM-PDA are also presented. To compare the performance of IMM-PDA algorithm, we implement the single KF-PDA models and the IMM-PDA approach. The experiment results demonstrate that the IMM-PDA not only catch the maneuver but also can give the best estimation results.

Actually, there is another big challenge for hand tracking in the gesture study community, i.e., accurately detecting/measuring the hand position and velocity at each frame. Since the current work is a postprocess to deal with the VCM output data, it can not feedback information to help VCM. So far, we still can not successfully estimate the hand state in the long (tens of thousands frames) video sequences, because the VCM measurements are too noisy to be handled by the filters in some segments. One idea is to combine IMM-PDA with VCM, and do online tracking. Another is to compare the PDA with other data association approaches, such as the *optimal Bayesian filter* [20], the *Joint Probabilistic Data Association approach* [21], the *multiple hypothesis tracking* [22], and the *interacting multipattern probability and joint probability data association approach* [23]. These should be our future work.

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